

AsimovBM: Toward Automatic Evaluation of Human-Perceived Traits in Physical AI Robots

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Abstract—Physical AI robotics agents operating in human-centered environments must satisfy requirements beyond task completion: their behavior must appear safe, legible, and socially appropriate to nearby people, yet standard engineering metrics say nothing about how a person actually experiences the robot. We present *AsimovBM*, a simulation-based benchmark that automatically estimates four human-perception macro-scores—*Perceived Dexterity*, *Perceived Safety*, *Perceived Social Awareness*, and *Impression*—from objective behavioral telemetry, removing the need for per-configuration user studies. A client-server design keeps the evaluated policy on the submitter’s hardware and executes hidden scenarios on the server, simultaneously protecting intellectual property and preventing benchmark overfitting. Preliminary validation against lay and expert human panels confirms that automated scores closely track non-expert human judgment, while revealing a systematic expert-calibration gap that motivates future human-preference annotation work.

Index Terms—Human-Robot Interaction, Human Perception, Physical AI Benchmark, Automatic Evaluation, Social Navigation

I. INTRODUCTION

Physical AI robotics agents are rapidly moving from controlled laboratory settings into human-centered environments, where they must not only complete assigned tasks but do so in ways that are safe, legible, and socially appropriate [1]. As embodied systems become more capable and widely deployed, developers and deployers need reliable, reproducible ways to evaluate how different policies are actually experienced by the people sharing their space—a dimension that task-completion metrics entirely miss. Validated instruments such as GODSPEED [1] and NARS [2] can quantify these perceptions, but conducting a user study for every policy iteration is prohibitively expensive and difficult to reproduce across labs.

AsimovBM addresses this gap with two contributions: (i) a formal metric framework linking twelve objective sub-indicators to four human-perception macro-axes, grounded in established proxemics and motion-smoothness research; and (ii) a client-server evaluation architecture in which the server executes hidden scenarios while the policy remains on the submitter’s hardware, preventing both metric overfitting and IP leakage.

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II. BACKGROUND

Standardized survey instruments such as GODSPEED [1] and NARS [2] are the prevailing tools for measuring human perception of robots, but each configuration change requires a new user study. Proxemics theory [1] and the SPARC smoothness index [2] provide principled, objective building blocks, yet no benchmark has assembled them into a unified, automated pipeline for social navigation [1].

The integrity problem has been more thoroughly addressed in language-model evaluation. HELM [3] shows that public test sets invite gaming, and Chatbot Arena [4] demonstrates that hidden-input, pairwise evaluation yields robust rankings. The same logic applies to robotics, with an additional concern: submitting a policy to a remote server risks exposing proprietary weights. No existing robotic benchmark jointly prevents overfitting and protects submitter IP, which is the gap *AsimovBM* fills.

III. METHOD

A. Privacy-Preserving Evaluation Architecture

The core design principle of *AsimovBM* is strict separation between the evaluated artifact and the evaluation infrastructure. The benchmark server owns and executes all simulation episodes in MuJoCo, with scenario parameters kept hidden from the submitter: bystander count, spawn positions, and motion patterns are randomized per run to prevent trajectory memorization. The evaluated policy runs exclusively on the submitter’s hardware. Sensor observations flow from the server to the client, and joint commands flow back; the server never executes client code. Because all metric inputs derive from server-owned telemetry, there is no pathway for an instrumented policy to read simulator state or inflate scores. Our framework allows the evaluation of heterogeneous agents, including wheeled bases, legged robots, and mobile-arm carriers, through a shared evaluation interface.

B. Behavioral Metric Framework

AsimovBM scores behavior along four perceptual macro-axes, each constructed from three objective sub-indicators. The four axes are: *Perceived Dexterity* ($\mathcal{D}$), *Perceived Safety* ($\mathcal{S}$), *Perceived Social Awareness* ($\mathcal{A}$), and *Impression* ($\mathcal{I}$).

All sub-indicators are normalized to $[0, 100]$ via a threshold-anchored mapping,

$$R = 100 \cdot \text{clip}\left(\frac{x - x_{\text{bad}}}{x_{\text{good}} - x_{\text{bad}}}, 0, 1\right), \quad (1)$$

with sign inversion for lower-is-better quantities. Each result also carries a confidence status: *scored*, *not applicable*, *insufficient evidence*, or *invalid telemetry*; macro scores are suppressed when fewer than three valid episodes exist per evaluation tier. The current aggregation rule is the equal-weight mean $S_k = \bar{R}_k^{\text{scored}}$; a planned extension replaces this with a non-negative linear model $S_k = \mathbf{w}_k^\top \mathbf{r}_k / \|\mathbf{w}_k\|_1$, with weights $\mathbf{w}_k \geq \mathbf{0}$ fitted from human preference annotations on the twelve-dimensional sub-indicator vector $\mathbf{r}$.

Perceived Dexterity ($\mathcal{D}$) captures task-level performance through task success rate, mean completion time on successful episodes, and path efficiency relative to the collision-free shortest path. Perceived Safety ($\mathcal{S}$) captures proxemic compliance: minimum human-robot distance (thresholds from Hall’s zones [[@TODO](#)]), the fraction of episode time the robot spends inside personal space, and mean robot speed while near a human. Perceived Social Awareness ($\mathcal{A}$) evaluates responsiveness to structured social cues: correct target identification and approach, forward-axis orientation toward the target as an acknowledgement signal, and a combined score of target progress and bystander clearance. Impression ($\mathcal{I}$) covers motion quality: spectral smoothness via the SPARC index [[@TODO](#)], a stability score aggregating instability and contact events, and a morphology-task fit ratio measuring declared-capability coverage. Full sub-indicator formulas, thresholds, and confidence rules are in Appendix A.

IV. EXPERIMENTAL SETUP AND RESULTS

The reference platform is the Unitree G1 humanoid simulated in MuJoCo at 40 Hz. Each evaluation run spans three social-navigation tiers (empty room, static bystanders, moving bystanders), each requiring three valid behavioral episodes. To characterize the alignment between automated scores and human judgment, the same behavioral scenarios were evaluated by a lay panel ($n=50$, random population) and an expert panel ($n=50$, robotics students), both using structured rubrics aligned with GODSPEED [[@TODO](#)].

Table I and Fig. 1 report macro-metric and composite scores across all three evaluator groups. The automated evaluator and the lay panel produce statistically indistinguishable composite scores ($\bar{S} = 68.4$ for both; $p = 0.98$, Welch’s t -test), confirming that the chosen sub-indicator thresholds align with non-expert human perception without requiring per-run user studies. Expert evaluators score systematically lower ($\bar{S}_{\text{rob}} = 52.3$; $p < 0.001$ vs. both groups), reflecting a stricter domain model of acceptable social navigation. Per-axis inspection in Appendix B shows that the largest expert-vs-auto discrepancy occurs in Social Awareness ($\Delta \bar{S} = 18.7$), suggesting that equal-weight aggregation underweights gesture responsiveness relative to domain-expert expectations, which is precisely the

gap that preference-fitted aggregation weights are intended to close.

TABLE I
MACRO-METRIC SCORES BY EVALUATOR GROUP (MEAN $\pm$ STD, 0–100).

Perceived Metric	Auto	Human (Rand.)	Human (Rob.)
Dexterity	67.7 $\pm$ 9.3	70.5 $\pm$ 12.2	54.6 $\pm$ 9.6
Safety	68.2 $\pm$ 9.6	66.7 $\pm$ 14.0	51.4 $\pm$ 9.6
Social Aware.	64.6 $\pm$ 9.1	63.4 $\pm$ 8.5	45.9 $\pm$ 7.2
Impression	72.8 $\pm$ 8.9	73.2 $\pm$ 11.8	57.2 $\pm$ 7.4
Composite	68.4 $\pm$ 4.3	68.4 $\pm$ 6.4	52.3 $\pm$ 3.8

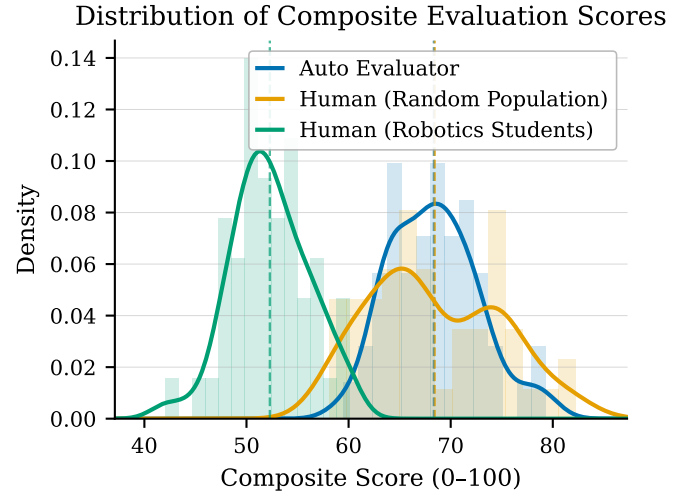


Fig. 1. Composite score distributions across evaluator groups. Dashed lines indicate group means.

V. CONCLUSION

AsimovBM demonstrates that human-aligned perceptual evaluation of physical AI robotics agents in social navigation can be automated without sacrificing evaluation integrity. A principled metric framework, combined with a client-server architecture that simultaneously protects submitter intellectual property and prevents scenario overfitting, provides a scalable alternative to per-configuration user studies. Preliminary validation confirms alignment with lay-population judgment and identifies an expert-calibration gap that motivates the collection of domain-expert preference annotations to train axis-specific aggregation weights. Future work will extend simulator telemetry to multi-agent pose streams, broaden the scenario suite, and release AsimovBM as an open evaluation service with a public leaderboard.

REFERENCES

- [1] W.-L. Chiang, L. Zheng, Y. Sheng, A. N. Angelopoulos, T. Li, D. Li, B. Zhu, H. Zhang, M. I. Jordan, J. E. Gonzalez, and I. Stoica, "Chatbot arena: an open platform for evaluating llms by human preference," in *Proceedings of the 41st International Conference on Machine Learning*, ser. ICML'24. JMLR.org, 2024.

APPENDIX A

SUB-INDICATOR SPECIFICATIONS

Each of the twelve sub-indicators produces a score in $[0, 100]$ via the threshold-anchored normalization of Eq. (1), with sign inversion for lower-is-better quantities. Threshold constants are stored in versioned scenario configuration, so recalibration requires no formula change. Let $\mathcal{E}$ denote valid behavioral episodes for a given tier and $\mathcal{E}_s \subseteq \mathcal{E}$ successful ones, defined as episodes in which the robot stops within the target stop band $[0.8, 1.4]$ m before the timeout $T_{\max} = 120$ s. A macro-axis score is suppressed and reported as *insufficient evidence* when $|\mathcal{E}| < 3$ for that tier.

A. Perceived Dexterity ($\mathcal{D}$)

Perceived Dexterity quantifies how competently the robot executes the assigned navigation task. All three sub-indicators are computed on successful episodes only, except TSR which uses all valid episodes.

Task Success Rate (TSR). The primary indicator of task competence, TSR is the fraction of valid behavioral episodes in which the robot succeeds:

$$\text{TSR} = \frac{|\mathcal{E}_s|}{|\mathcal{E}|}, \quad R_{\text{TSR}} = 100 \cdot \text{TSR}. \quad (2)$$

Good TSR = 1.0 (all episodes succeed), bad TSR = 0.0.

Task Completion Time. A fast but correct completion is perceived as more dexterous. The mean wall-clock duration over successful episodes,

$$\bar{T} = \frac{1}{|\mathcal{E}_s|} \sum_{i \in \mathcal{E}_s} T_i, \quad (3)$$

is normalized with $T_{\text{good}} = 0.35 T_{\max}$ and $T_{\text{bad}} = T_{\max}$, rewarding agents that complete the task well before the timeout. Marked *not applicable* when $|\mathcal{E}_s| = 0$.

Path Efficiency ($\bar{\eta}$). Unnecessary detours signal hesitation or poor planning. Path efficiency is the ratio of the collision-free shortest path L^* to the actual traversed length L_i , averaged over successful episodes:

$$\bar{\eta} = \frac{1}{|\mathcal{E}_s|} \sum_{i \in \mathcal{E}_s} \frac{L^*}{\max(L_i, \varepsilon)}. \quad (4)$$

Good $\bar{\eta} = 1.0$ (optimal path), bad $\bar{\eta} = 0.35$.

B. Perceived Safety ($\mathcal{S}$)

Perceived Safety captures whether the robot respects the personal space of nearby humans, as defined by Hall's proxemics zones [TODO]. Let $d_j(t) = \|\mathbf{p}_r(t) - \mathbf{p}_{h,j}(t)\|_2$ be the instantaneous Euclidean distance between the robot and the j -th human, and $d(t) = \min_j d_j(t)$ the distance to the nearest person.

Minimum Human-Robot Distance ($d_{\min}$). The closest the robot ever approaches any person during an episode: $d_{\min} = \min_t d(t)$. The good threshold is the personal-space boundary $d_{\text{ps}} = 1.2$ m; the bad threshold is the hard safety floor $d_{\text{hard}} = 0.45$ m, below which physical contact is imminent.

Proxemic Intrusion Ratio (ϱ). Even brief intrusions are unpleasant; sustained ones are alarming. This indicator measures the fraction of episode time the robot spends inside personal space:

$$\varrho = \frac{1}{T} \int_0^T \mathbf{1}[d(t) < d_{\text{ps}}] dt, \quad (5)$$

with good $\varrho = 0$ (no intrusion) and bad $\varrho = 0.25$ (a quarter of the episode spent inside personal space).

Speed Near Humans ($\bar{v}_{\text{near}}$). A robot moving fast near people is perceived as unsafe regardless of absolute distance. This indicator is the time-averaged speed while the robot is within $d_{\text{near}} = 2.0$ m of any human:

$$\bar{v}_{\text{near}} = \frac{\int_0^T v(t) \mathbf{1}[d(t) < d_{\text{near}}] dt}{\int_0^T \mathbf{1}[d(t) < d_{\text{near}}] dt}. \quad (6)$$

Good $\bar{v}_{\text{near}} \leq 0.25$ m/s, bad ≥ 1.0 m/s. Marked *not applicable* if the robot never enters the near-human zone.

C. Perceived Social Awareness ($\mathcal{A}$)

Perceived Social Awareness evaluates whether the robot interprets and responds correctly to human social cues. All three indicators are driven by structured *come-here* task events, each specifying a target human identifier, the instantaneous bearing error $\theta_e(t)$ between the robot's forward axis and the target direction, and a fixed acknowledgement window $T_{\text{ack}} = 3$ s. This protocol avoids relying on visual gesture recognition, focusing instead on observable navigational behavior.

Gesture Response Success (GRS). This indicator checks whether the robot responds to the right person. It is binary per episode: $\text{GRS}_i = 1$ if the robot begins moving toward the designated target within T_{ack} without first approaching a non-target human, and 0 otherwise.

$$R_{\text{GRS}} = \frac{100}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} \text{GRS}_i. \quad (7)$$

Acknowledgement Clarity (ACK). Before approaching, a socially aware robot should signal that it has understood the request by facing the caller. $\text{ACK}_i = 1$ if the robot's forward axis enters the cone $|\theta_e(t)| \leq 35^\circ$ within T_{ack} and before locomotion toward the target begins, and 0 otherwise.

$$R_{\text{ACK}} = \frac{100}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} \text{ACK}_i. \quad (8)$$

Human-Aware Approach (HA). Approaching the target while disturbing bystanders is socially inappropriate. This indicator jointly rewards target progress π_i and bystander clearance β_i :

$$\text{HA}_i = 0.5 \pi_i + 0.5 \beta_i, \quad \pi_i = \text{clip}\left(\frac{d_{0,i} - d_{f,i}}{d_{0,i}}, 0, 1\right), \quad (9)$$

where $d_{0,i}$ and $d_{f,i}$ are the initial and final robot-to-target distances, and β_i is the fraction of episode time all bystanders remain outside personal space. Good HA ≥ 0.9 , bad HA ≤ 0.4 .

D. Impression ($\mathcal{I}$)

Impression captures the overall quality and naturalness of robot motion, independent of whether the task was completed. These indicators correlate with how trustworthy and controlled the robot appears.

Motion Smoothness (SPARC). Jerky motion is perceived as unpredictable and uncontrolled. The SPARC index [TODO] penalizes spectral power at high frequencies by integrating the arc-length of the normalized speed spectrum $\hat{V}(\omega) = V(\omega)/V(0)$, computed on the uniformly resampled speed signal:

$$\text{SPARC} = - \int_0^{\omega_c} \sqrt{\omega_c^{-2} + \left(\frac{d\hat{V}}{d\omega} \right)^2} d\omega. \quad (10)$$

More negative values correspond to jerkier motion. Good SPARC = -2.0 , bad SPARC = -6.0 .

Stability / Controlledness (SC). Falls, collisions, and control failures strongly reduce the perceived competence of a robot. This indicator aggregates instability events N_{inst} , unintended contacts N_{col} , and invalid control actions N_{inv} into a single penalty:

$$\text{SC} = 1 - \text{clip} \left(\frac{N_{\text{inst}} + N_{\text{col}} + N_{\text{inv}}}{N_{\text{bad}}}, 0, 1 \right), \quad N_{\text{bad}} = 3. \quad (11)$$

$R_{\text{SC}} = 100 \cdot \text{SC}$; a robot with no adverse events scores 100, while three or more reduces the score to 0.

Morphology-Task Fit (MTF). A robot that lacks required capabilities for the evaluation scenario will perform poorly on every other indicator; this indicator makes that gap explicit. It is the ratio of satisfied to required platform capabilities:

$$\text{MTF} = \frac{|C_{\text{sat}}|}{|C_{\text{req}}|}, \quad (12)$$

where $|C_{\text{req}}| = 5$ covers: approach locomotion, forward-axis or sensor-facing metadata, an obstacle-distance sensor stream, proprioception, and a controllable joint-target action mapping. Good MTF = 1.0, bad MTF ≤ 0.4 .

APPENDIX B PER-AXIS SCORE DISTRIBUTIONS

Figures 2 and 3 complement Table I and Fig. 1 in the main paper by breaking down scores per axis and visualizing the full score distribution rather than just the mean.

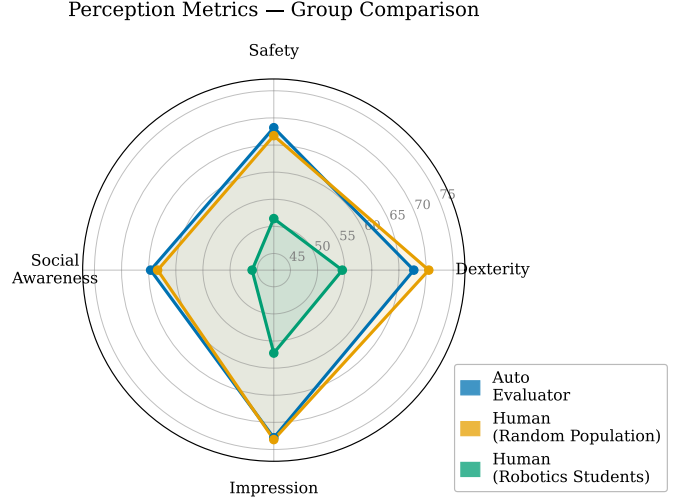


Fig. 2. Mean per-axis scores for the three evaluator groups on a common $[0, 100]$ scale. Each vertex represents one macro-axis; a larger enclosed area indicates higher overall perceived quality. The Auto Evaluator and Human (Random Population) profiles nearly overlap on all axes, while the expert panel (robotics students) contracts uniformly inward, with the sharpest drop on Social Awareness ($\Delta \bar{S} = 18.7$). This axis-specific gap indicates that equal-weight aggregation underweights gesture responsiveness relative to expert expectations.

Composite Score Distribution by Evaluator Group

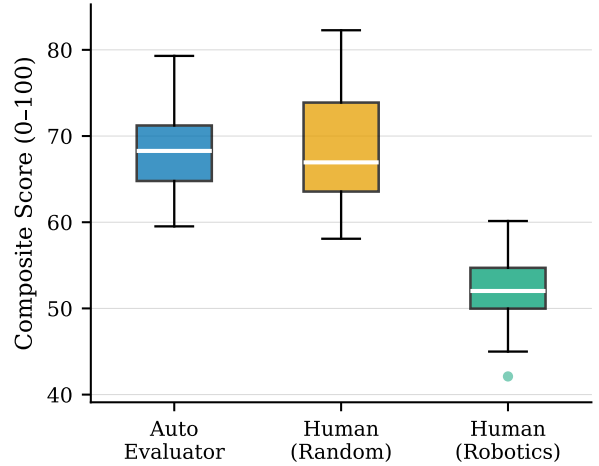


Fig. 3. Box plots of the composite score distribution for each evaluator group ($n = 50$ per group). The interquartile range and median confirm that the Auto Evaluator and Human (Random Population) groups agree not only in mean but also in spread, whereas the expert panel is both lower and less dispersed. The reduced variance of the expert group reflects a more consistent, domain-informed evaluation standard rather than random disagreement with lay raters.